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Enhancing Anaerobic Digestion of Food Waste by Combining Carriers and Microaeration: Performance and Potential Mechanisms

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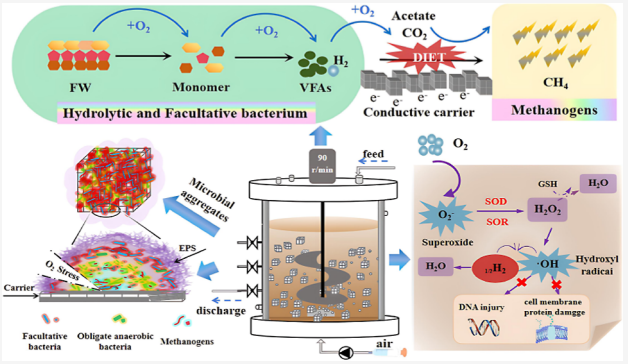
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ABSTRACT: Enhancing the anaerobic digestion (AD) of organic wastes has been a widely discussed topic. This study aims to enhance AD performance by combining microaeration with conductive or nonconductive carriers, using food waste as the substrate. The use of carriers alone enhanced methane production, and microaeration further improved performance. The conductive carrier showed significant enhancement with microaeration, achieving a daily methane yield of 478 ± 11.3 mL CH_4/g VS, which was 1.1 and 1.3 times higher than that of the nonconductive carrier and the control digester, respectively. Furthermore, the study explored various aspects, including oxidative stress, antioxidant capacity, and microbial community structure during digestion. The results demonstrated that combining microaeration with a conductive carrier improved hydrolytic-acidification efficiency and promoted direct interspecies electron transfer among syntrophic microorganisms. Methanogenic archaea aggregated on the carrier surface and formed consortia with facultative anaerobes, thereby mitigating oxidative stress effects on cells and enhancing total methane production. Moreover, metabolomics analysis showed that combining conductive carriers with microaeration enhanced ATP transport across the bacterial membrane, accelerated nutrient conversion, and caused significant changes in metabolites and intermediates related to glycerophospholipids, amino acids, and signal transduction pathways.



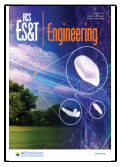
KEYWORDS: *anaerobic digestion, microaeration, carrier, oxidative stress, metabonomics*

1. INTRODUCTION

In recent years, the management of municipal solid waste (MSW) has become a significant global environmental challenge. In 2019, over 2 billion tonnes of MSW were produced globally, with projections suggesting a potential 70% increase by 2050.[1] The China Statistical Yearbook (2020) reported that MSW production in China has steadily risen at a rate of 5−6% annually, reaching 254.79 million tons in 2020, while food waste (FW) production in China ranged between 72.44 million to 127.40 million tons, comprising 30−50% of all MSW in the same year.[2] FW contains valuable carbohydrates, proteins, lipids, and minerals, making it an ideal raw material for biofuel production via microbial conversion.[3] Anaerobic digestion (AD) is a key process in the microbial conversion of FW. During digestion, fermenting bacteria and methanogenic archaea work together to convert organic compounds in FW into methane and carbon dioxide. This process involves stages such as hydrolysis, acidification, hydrogen-producing acetogenesis, and methanogenesis. Methane, as a clean fuel, has significant applications in the energy supply and industrial production. Therefore, enhancing the efficiency of converting FW organics to methane is critical. Various methods have been employed to enhance the hydrolytic-acidification efficiency, including pretreatments (e.g., mechanical, thermal, and acid/alkali),[4] microaeration,[5,6] and the use of exogenous additives.[7,8]

Microaeration involves injecting small amounts of air or oxygen (typically ranging from 0.005 to 5 L O_2/L reactor/day) into an anaerobic system, and it has garnered considerable research attention for its potential to improve the hydrolytic-acidification process in the digestion of various organic substrates.[9−11] Injecting small amounts of air creates a higher

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oxidation−reduction potential (ORP) environment compared with strict anaerobic conditions, thereby reshaping the microbial community in the digester. Facultative anaerobes involved in the hydrolytic-acidification process exhibit higher activities under these conditions. Fu et al.[5] demonstrated that introducing microaeration at an air load of 12.5 mL/L_R per day in the AD of corn stover enhanced the biodiversity of facultative hydrolytic and acidifying bacteria. Furthermore, according to Yu et al.,[12] appropriate microaeration (5 mg O_2/L-reactor/cycle) not only enhanced the hydrolytic-acidification process in the digestion of black water but also improved the methanogenesis efficiency. The ORP variation induced by microaeration also had domestication effects on the archaeal community. Higher abundance of hydrogenotrophic methanogens with stronger toxic tolerant ability was commonly observed in the microaerated digesters.

In addition, the syntrophic interaction between bacteria and methanogenic archaea is another crucial factor determining the AD efficiency.[13] These microorganisms have unique yet complementary metabolic capabilities, exchanging electrons via H_2 as electron shuttles for energy. Establishing direct interspecies electron transfer (DIET) between bacteria and archaea is essential for efficient electron communication. DIET established by exogenous conductive materials enhances both the transfer rate and efficiency by building an electron bridge between syntrophic oxidation bacteria and methanogens while also avoiding mass-transfer resistance through interspecies hydrogen transfer. Thus, the combination of microaeration and conductive materials holds promise for achieving significant improvements. Our previous research has shown that combining CaO_2 (0.01 g/L) and magnetite (0.4 g/L) synergistically enhanced the conversion of organic matter, optimizing the relationship between fermentative bacteria and methanogenic archaea to increase methane production overall.[14] The release of O_2 or H_2O_2 during CaO_2 decomposition can be considered another form of microaeration. Additionally, microaeration introduces reactive oxygen species (ROS) into the reactor, resulting in their accumulation, inducing toxic oxidative stress in cells or organisms. This stress can lead to oxidative damage to microbial activity (or central metabolism) and important cellular molecules (e.g., DNA damage and protein inactivation).[15,16] Therefore, given the toxicity of these oxidative agents to anaerobic microbes, precise control of oxygen dosage is crucial to prevent oxidative stress on strict anaerobic functional communities, especially oxygen intolerant methanogenic archaea. Furthermore, the substantial biomass loss during feeding and discharging exacerbated challenges for anaerobic microorganisms coping with oxidative stress, particularly slow growing methanogens that could be easily screened out during sludge discharge, leading to system failures in long-term microaeration applications.

Introducing carriers like zeolite,[17] nonwoven fabric[18] and fibers[19] into the digestive system can effectively retain slow growing anaerobic microorganisms, reducing excessive loss of functional microorganisms during semicontinuous AD operation with daily feeding and discharging. Carriers with conductive properties were preferred because they not only induced anaerobic biofilm formation but also promoted the DIET mechanism. Liu et al.[19] discovered that gaps and pores in fibrous carriers facilitated anaerobe attachment during corn straw AD, enabling easy contact between microorganisms and the substrate. Zhuravleva et al.[20] observed that biofilms were more abundant on carbon cloth than on fiberglass cloth in a continuous flow bioreactor, and using carbon cloth improved DIET between electrogenic and electrotrophic microorganisms. Microbial biofilms consist of cells embedded in a three-dimensional extracellular matrix that acts as a scaffold for cell adhesion, supports mature biofilm structure formation, and protects cells from harsh environmental conditions.[21−23] From the aspect of oxygen toleration for microbes, the formation of biofilm on the carrier's surface may present a certain protection on strict anaerobes.[15]

Based upon analysis, microaeration has been shown to enhance the hydrolytic-acidification process. Additionally, conductive carriers may facilitate efficient electron communication between archaea and bacteria while protecting strict anaerobes from oxidative stress. Therefore, combining microaeration and conductive carriers in an AD system is expected to greatly improve the digestive performance. No relevant reports have been found to date. Thus, this study aimed to regulate the digestive system in semicontinuous AD of FW by combining microaeration and conductive carriers. Specifically, our objectives were: (1) to improve hydrolytic-acidification efficiency to supply adequate substrates for methanogenic archaea and (2) to establish effective electron communication and metabolic balance between bacteria and archaea to ensure efficient utilization of bacterial metabolic products and maintain system stability. To achieve these objectives, the study initially analyzed digestive performance to assess the promotional effect. Cell viability analysis, enzyme activity measurement, and microbial community analysis were used to understand microbial metabolic activities under such artificial conditions. Furthermore, metabolomics analysis was employed to elucidate the internal mechanisms of microorganisms. The results of this study may provide new insights and references for further improving the AD efficiency.

2. MATERIALS AND METHODS

2.1. Substrates and Inoculum. The anaerobic inoculum was sourced from a wastewater treatment plant in Chongqing, China. The substrate FW was prepared in the laboratory according to the formula described by Li et al.[24] Characteristics of both the substrate and inoculum are detailed in Table S1.

2.2. Digester Setup and Operational Conditions. The AD system comprises a feed hopper, a mixing shaft, a separate jacketed thermostat, and a gas collection device. Each reactor was constructed from a plexiglass cylinder (inner diameter 8 cm, height 25 cm) with a working volume of 0.7 L and a headspace of 0.3 L. The carrier filling ratio in this study was 20%, determined as optimal from preliminary experiments (Figure S1). The experimental group with added carriers included two reactors: one containing carbonaceous honeycomb-like carrier (noted as T20) (2 cm × 2 cm × 2 cm, Suzhou Carbon whirlwind Carbon Co. Ltd., China) and the other packed with polyurethane sponge without conductivity (noted as D20) (2 cm × 2 cm × 2 cm, Henan Tianyi Environmental Protection Science and Technology Co. Ltd., China). Detailed characteristics of the carriers can be found in Table S2. The digester without a carrier served as the control group (noted as D0). The AD systems operated in the semicontinuous feeding mode to maintain a constant hydraulic retention time of 20 d, with an organic loading rate of 2.5 g VS/L·d and a digestion temperature of 35 ± 1 °C. The entire experimental process comprised three stages: initiation (stage I), nonaeration (stage II), and microaeration (stage III). In the third stage, air was injected into the digesters at a dose of 5 mL

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using a syringe at a uniform and slow rate after feeding, leading to an increase in the ORP by 30−50 mV. All of the reactors were carried out in triplicate.

2.3. Chemical and Physical Analysis. Measurements of total solid (TS), volatile solid (VS), chemical oxygen demand, and NH_4^+-N were conducted in accordance with standard methods.[25] pH was determined by a pH meter (FE28, METTLER TOLEDO, Switzerland). Conductivity was measured by a conductivity meter (DDS-11A/307, Yueping, Shanghai). The biogas composition and volatile fatty acids (VFAs) were determined by gas chromatography (GC1690, Jiedao, Hangzhou). The operation process of gas chromatography was introduced in detail in previous study.[26] The extraction method of extracellular polymeric substances (EPS) referred to the study of Li et al.[27] Based on the difficulty of component and bacterial separation as well as spatial distribution, EPS was subdivided into three layers: soluble EPS (SEPS), loosely bound EPS (LB-EPS), and tightly bound EPS (TB-EPS). The PN and PS of each form of EPS were determined as the total amount by the Lowry−Folin method and phenol-sulfuric acid method, respectively.[28] According to Limbach et al.,[29] the intracellular ROS production was determined using 2′7′-dichlorodihydrofluorescein diacetate (H2DCFDA; Molecular Probes, Eugene, USA). Electron transfer activity (ETS) was measured using the method reported by Luo et al.[30] α-Glucosidase activity, ATP content, catalase (CAT), superoxide dismutase (SOD), superoxide reductase (SOR), nicotinamide adenine dinucleotide (NAD^+/NADH), as well as acetate kinase (ACK) were measured by using the ELISA kit (Shanghai hengyuan biological technology co., LTD).

2.4. Microbiological Analysis. Anaerobic biofilms on the surface of the carrier and digested sludge were detected by scanning electron microscopy (SEM, Hitachi S-4800, Leica, Germany). The presence of a carrier may create a physical barrier or provide an attachment point, thereby impeding or enhancing the contact of the colorant with the microorganisms. Therefore, in this study, sludge samples were collected for staining to assess the effectiveness of carriers in enhancing cell viability and optimizing microbial activity during AD after the introduction of microaeration. The cell viability of the sludge samples was stained using Live/Dead BacLight bacterial viability staining kit (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's instructions and observation using a confocal laser scanning microscope (ZEISS Axio Scan, Z1, Germany). Fluorescence in situ hybridization (FISH) analysis was performed on biomass samples fixed with 4% paraformaldehyde solution.[31] This study utilized oligonucleotide probes specific for bacteria (FITC-labeled EUB338mix probes) and archaea (Cy3-labeled ARC915 probe). The labeled samples were visualized using an Olympus BX41 epifluorescence microscope (Olympus, USA) (Table S3). At the end of each stage, samples were taken from the digester for microbial community structure analysis. To explore the differences in bacterial metabolic characteristics, six biomass samples were collected from each reactor for metabolite analysis at the end of the experiment to ensure the reliability of the experimental results. The details of the analytical methods used above are available in the Supporting Information.

2.5. Statistical Analysis. The experimental data underwent analysis of variance (ANOVA) for statistical processing. All physicochemical index data presented in this paper are expressed as "mean ± standard deviation" (n = 3) to enhance clarity. Student's t-test was used to assess significant differences between different systems, with a p-value below 0.05 considered statistically significant. Statistical mapping was performed by using Origin 9.0 software.

3. RESULTS AND DISCUSSION

3.1. Anaerobic Digestion Performance. The digestion performance of various digesters is depicted in Figure 1.

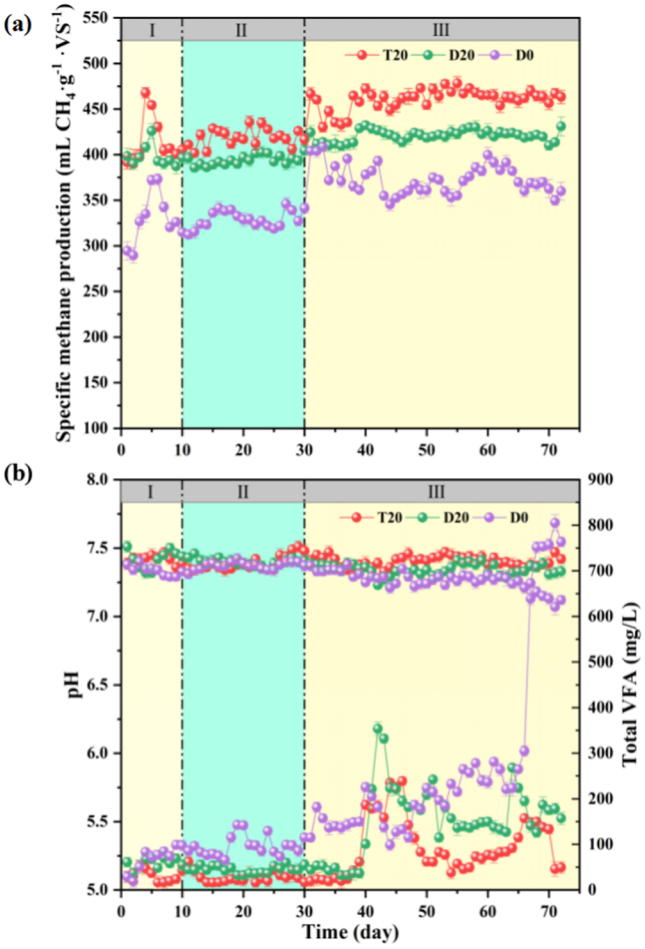


Figure 1. (a) Specific methane production and (b) TVFA and pH in different digesters.

Methanogenic properties are commonly believed to directly reflect the adaptability of the anaerobic microbiome in AD.[32] Figure 1 shows that methane production achieved stable after an short time start and varies among different digesters. Clearly, the application with carrier materials significantly enhanced methane production (p < 0.05). The averaged methane yield for the control digester was approximately 329 ± 19.1 mL CH_4/g VS, whereas for the D20 and T20 in this stage were 394 ± 12.5 and 420 ± 17.9 mL CH_4/g VS, respectively. These results indicated that carrier materials positively enhanced digestion performance, especially conductive carriers, which exhibited a greater promotional effect compared to nonconductive ones.[33]

Trace amount of air was added at the end of stage II (30 days). In this stage (stage III), it can be noted that the application of microaeration gives a further promotion of the methane production. For the control digester, the methane production was increased from 329 ± 19.1 mL CH_4/g VS at

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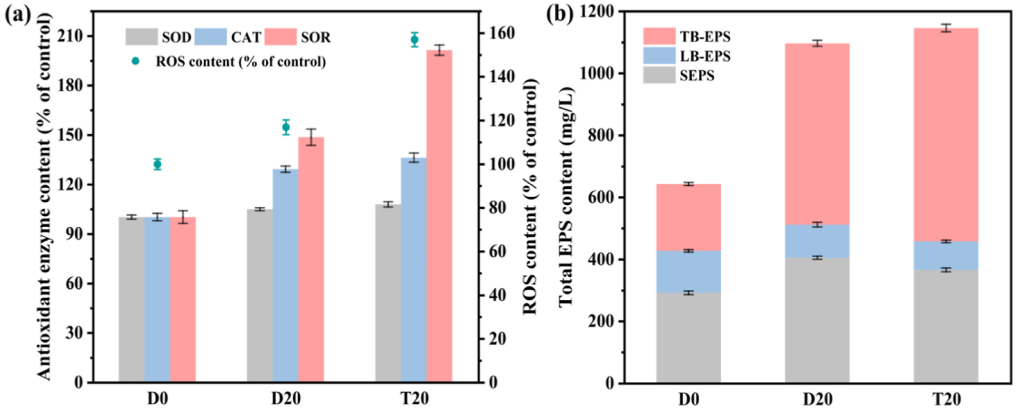


Figure 2. (a) ROS production and secretion of antioxidative enzymes and (b) EPS content in stage III.

stage II to 368 ± 15.9 mL CH_4/g VS. For the digester D20, the methane production increased from 394 ± 12.5 mL CH_4/g VS to 430 ± 7.1 mL CH_4/g VS. T20 presented the highest methane production with the value of 478 ± 11.3 mL CH_4/gVS, which was 1.3 times higher than that of the control digester at this stage. By comparing the data before and after microaeration, an interesting phenomenon could also be observed that microaeration promoted conductive carriers more effectively than other methods, indicating a more significant enhancement effect when combining microaeration with conductive carriers ($P < 0.05$).[14]

VFAs, such as acetate, propionate, and butyrate, are important intermediate products in the AD process. Accumulation of VFAs in the digester can lower the pH and inhibit methane production. Following the introduction of microaeration (stage III), D0 exhibited significant fluctuations in specific methane yield and TVFA levels, with a noticeable trend toward accumulation over time (Figure 1b). A rapid increase in TVFA concentration was observed in this digester after 35 days of digestion following the introduction of microaeration, accompanied by decreased methane production and pH, suggesting system acidification. This result verified our assertion in the introduction that the application of microaeration needs to be well controlled to avoid its potential damage to obligate anaerobes. However, when the microaeration is combined with carriers, the system stability was much more enhanced. Despite inevitable fluctuations in VFA concentrations, they consistently remained below 300 mg/L throughout the experimental period. Furthermore, the VFA concentration of the T20 was much lower than that in the D20, possibly due to enhanced VFA consumption facilitated by the conductive material.[34] Overall, combining carriers with microaeration mitigates its adverse effects on obligate anaerobes, especially for some methanogens, which is helpful to maintain the system stability. Moreover, the combined use of conductive carrier and microaeration demonstrated a more favorable effect.

3.2. Oxidative Stress and Antioxidant Ability.

Exposure of microorganisms to oxygen leads to the accumulation of molecular oxygen within the cell, resulting in the production of large quantities of ROS (i.e., $O_2^{\bullet-}$, H_2O_2, and $^{\bullet}OH$), which can cause oxidative stress, enzyme inactivation, and interference with microbial morphology and structural integrity.[35] Figure 2 showed the ROS intensities in different digesters, indicating that the use of carrier materials improves the ROS intensity. The ROS intensity in D20 with the nonconductive polyurethane sponge was 117% higher than the control digester with microaeration solely. Conductive carriers promoted ROS intensity even further, with an increase of 157% compared to the control. The reason for this is that polymer synthetic hydrophilic biological carrier, characterized by interconnected pores that increase contact area and time, allows oxygen to remain within the carrier. This promoted interactions between oxygen and microbial cells, while also modifying pH, temperature, and nutrient supply to create favorable conditions for ROS production.[36,37] The conductive carrier materials could promote electrochemical reactions within the digester,[38] leading to increased electron-transfer processes and further promoting the generation of ROS through mechanisms such as electron capture by oxygen. Notably, the higher quantity of microbes in the conductive material group may promote microbial proliferation and metabolic activity, thereby contributing to the elevated ROS levels in T20 (Figure 5). Additionally, conductive carriers were optimal catalysts for H_2O_2 production due to their high selectivity for the two-electron oxygen reduction reaction, good electrical conductivity, and nontoxicity.[39] ROS can serve as terminal electron receptors for anaerobic respiration and support the proliferation of facultative anaerobic bacteria, contributing to enhance hydrolytic-acidification processes and accelerate the conversion of particulate organic matter to dissolved ones.[40] In addition, ROS also has strong oxidizing ability, which can produce free radicals and help oxidize various refractory organics.[41,42]

Prolonged exposure of cells to ROS can induce cellular damage. In response, microorganisms can produce antioxidative enzymes like CAT, SOR, and SOD to neutralize harmful oxidants, converting them into harmless products and preventing damage to cellular components.[43] Specifically, SOD and SOR primarily mitigate damage caused by $O_2^{\bullet-}$ in anaerobic systems by reducing $O_2^{\bullet-}$ peroxyl radicals to H_2O_2. CAT catalyzes the decomposition of H_2O_2 into O_2 and H_2O, thereby reducing its potentially toxic effects on anaerobic microorganisms. As depicted in Figure 2a, following prolonged exposure to microaeration, the levels of SOD, SOR, and CAT in D20 were 104.8, 148.5, and 116.9%, respectively, compared to those in D0. Furthermore, in T20 exposed to microaeration, the levels of SOD, SOR, and CAT were further increased to 107.7, 201.4, and 136.1% of the control, indicating that conductive carrier enhanced the antioxidant capacity of the digester. Enhanced production of antioxidant enzymes helped microorganisms mitigate the adverse effects of excessive ROS

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and achieve redox balance, thereby protecting obligate anaerobes from oxygen stress.[44] Visual inspection of live/dead cells confirmed this finding, showing a greater proportion of live cells in the carrier group (Figure S2). Comparing D20 and T20 indicated that introducing conductive carriers with microaeration prompted microbes to adopt a distinct metabolic pattern in response to elevated ROS stress.

EPS, a polymer secreted by microorganisms in specific environments, envelops nearly all microbial cells, crucially facilitating biofilm formation, maintaining sludge agglomerate structure, and shielding microbial cells from adverse environmental conditions.[45−47] It was presumed that it may contribute to microbial aggregate formation and protect microorganisms from oxygen stress. Thus, EPS levels were quantified in the sludge following stable operation of each group (Figure 2b). Data revealed that SEPS predominated in the control group, whereas TB-EPS constituted the majority in carrier-containing sludge samples. SEPS were weakly bound with cells or absolutely dissolved into bulk solution,[48] suggesting that long-term microaeration could disrupt the original stable sludge structure, promote the conversion of other EPS forms into SEPS, and increase the possibility of sludge loss. This finding also elucidated why the carrier group exhibited superior operational effectiveness compared to that of the control group. TB-EPS plays an indispensable role in promoting sludge agglomeration and granulation by condensing cells and inert particles together through a solid physicochemical connection, resulting in a more compact and compacted sludge structure.[49] The contents of TB-EPS in the control group, D20, and T20 groups were 216.0, 586.3, and 689.5 mg/L, respectively. Furthermore, protein, an important component of EPS, had the highest content in T20, suggesting that microorganisms were stimulated by microaeration and enhanced the secretion of proteins (Table S4), which further promoted the content of TB-EPS and strengthened the adhesion of cells to each other, finally forming microbial aggregates. Microbial aggregates shield organisms living deep inside diffusion barriers, preventing the full penetration by oxygen.[50]

3.3. Enzymes and ETS Activities. AD is catalyzed by microorganisms, including hydrolytic acidification bacteria and methanogenic archaea, and involves synergistic actions of various enzymes. To assess how microaeration and carriers influence microbial activities, the activities of relevant enzymes and the ETS were analyzed (Figure 3). In stage II without microaeration, the presence of carriers elevated the activities of the relevant enzymes, indicating enhanced colonization of microorganisms and biofilm formation, leading to greater metabolic activity. Moreover, the inherent conductivity of the conductive carrier facilitated electron transfer, further enhancing enzyme activity and ETS. With microaeration introduction, enzyme activity and ETS further increased, indicating enhanced microbial activity. Subsequent discussion will focus on microaeration's promotion of enzyme activities.

α-glucosidase plays an important role in the hydrolysis of polysaccharides, facilitating the conversion of complex organic solids into soluble small molecules for microbial utilization.[51] The activity of α-glucosidase significantly increased after microaeration ($p < 0.05$), with the control group showing an increase from 266 IU/L to approximately 341 IU/L. Digesters with carriers showed greater improvements, reaching 401 IU/L (D20) and 494 IU/L (T20), which were 118% and 159% higher than those in the control digester. Similarly, microaeration resulted in an increase in the activity of ACK,

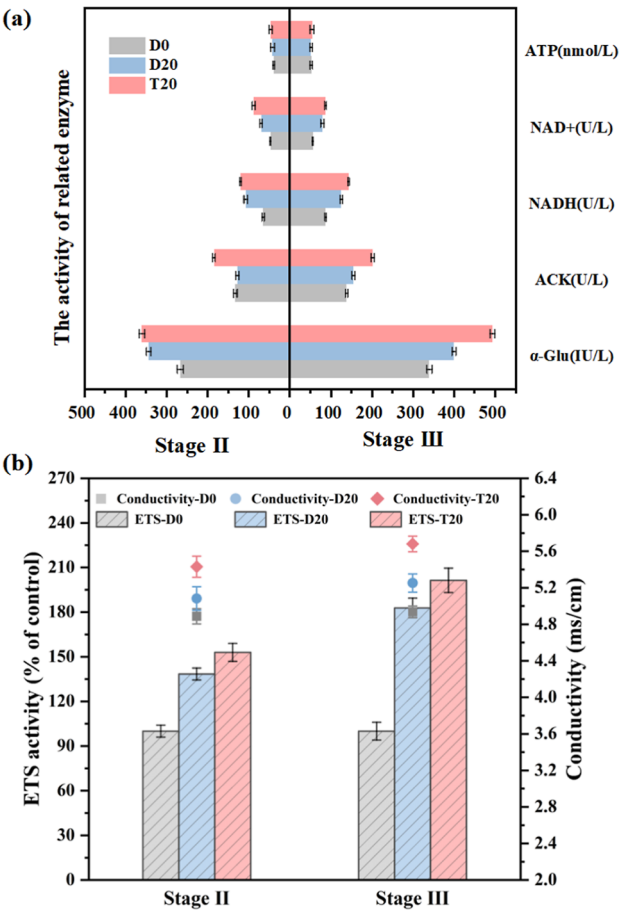


Figure 3. (a) Activity of related enzymes and (b) ETS and conductivity before and after microaeration treatment.

particularly in digesters with conductive carriers (184 to 202 U/L). These heightened activities of α-glucosidase and ACK provide evidence of the improvements brought about by microaeration and its synergistic combination with conductive carriers. Ruan et al.[51] observed that microaeration can effectively stimulate the activities of hydrolytic enzymes of facultative bacteria and enhance the activity of AD microorganisms. Furthermore, microaeration provided exogenous electron acceptors that facilitated the adjustment of bacteria from the reduction process with H^+ as the electron acceptor to the respiratory process with trace electron acceptors such as O_2, resulting in higher levels of energy metabolism.[14,52]

ATP, as the direct source of energy in living organisms, reflects cellular energy status and plays a critical role in regulating cellular metabolism.[53] During stage III, ATP activity exhibited a significant increase, with D20 and T20 digesters showing approximately 120% and 126% higher ATP activities compared to the control group. This increase in ATP activity may be attributed to microaeration stimulating the transmembrane transport of protons (significant enrichment of ABC transporter proteins) (Figure 7), thereby enhancing ATP synthesis. Moreover, as more acetyl-CoA flows into the tricarboxylic acid (TCA) cycle to produce ATP, more nicotinamide adenine dinucleotide (NADH) is produced.[54] Reduction of O_2 with NADH releases a significant amount of energy, creating a proton gradient that drives ATP synthesis and the transport of other metabolites.[9]

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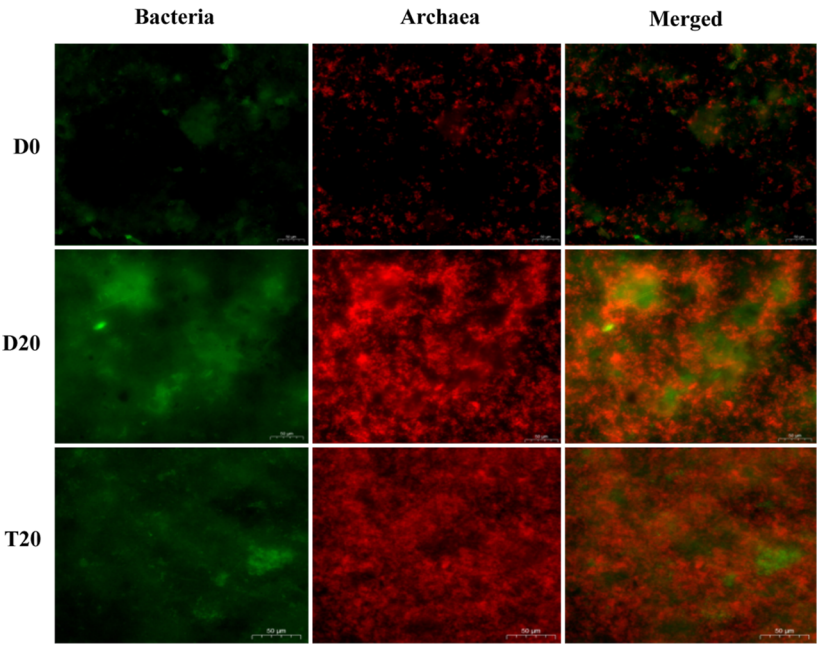


Figure 4. Spatial distribution of cells of archaea (Arc915-Cy3, red) and bacteria (EUB338mix-FITC, green) identified by FISH.

NAD (H/+) is a crucial cofactor in various biochemical processes within the cell.[55] NADH provides the necessary reduction equivalent for intracellular reduction, while NAD^+ serves as an electron acceptor in intracellular NAD^+ dependent oxidation reactions. Monitoring the $NADH/NAD^+$ ratio is critical in assessing the fermentation efficiency during the digestion process.[56] The addition of microaeration resulted in a slight increase in NAD^+ activity in T20, reaching 89.4 U/L, which was 154% and 111% higher than D0 and D20, respectively. Conversely, NADH activity significantly increased from 119.2 to 144.1 U/L, resulting in an approximately 15% higher ratio of $NADH/NAD^+$ ratio to 1.6. The lack of electron acceptors (e.g., oxygen) during the second stage may have inhibited the electron-transfer process and NAD^+ production.[57] Introducing microaeration stimulated the TCA cycle and enhanced NADH production. Furthermore, the increased activity of hydrolases after microaeration promoted the hydrolysis of polysaccharide organic matter, leading to the consumption of NAD^+ during glucose molecule catabolism[55] and consequently elevated $NADH/NAD^+$ values. High $NADH/NAD^+$ ratios positively regulated $O_2^{\bullet-}$ production,[58] reducing ROS induced stress.

ETS activity is an effective indicator of microbial and sludge biological activity. The introduction of microaeration had a slight impact on ETS activity in the control group but significantly improved activity in the carriers' utilized groups. The ETS activity of digester D20 increased from 138% to 183%, while T20 showed the highest ETS activity and conductivity, reaching 201% and 115% of the control group. These increases demonstrate enhanced electron transfer among different microorganisms.[59] In addition, conductive carriers established favorable electron-transfer pathways for microorganisms, promoting cellular electron transfer and metabolic processes in microorganisms by maintaining redox homeostasis.[14,60]

3.4. Microbial Analysis. *3.4.1. Visualization of Microbial Characteristics.* To visualize the impact of carriers on the aggregation of bacterial and archaeal cells, FISH analysis was conducted on suspended sludge (Figure 4). FISH images showed a notable overlap between red fluorescence from archaea and green fluorescence from bacteria in carrier groups, suggesting their spatial proximity. Semiquantitative analysis of fluorescence intensity indicated that archaeal fluorescence in the T20 and D20 groups constituted 93.1% and 80.5% of total bacterial fluorescence intensity, respectively (Figure S3). In contrast, the red fluorescence intensity in the control group was only 48.7% of the green fluorescence, suggesting a lower percentage of archaea compared with the carrier groups and even half that of the T20 group. These findings suggested that microorganisms in carrier-based samples were densely distributed, forming microbial aggregates, which highlighted carriers' role in facilitating microorganism attachment and protecting functional flora from adverse external conditions. This finding was also supported by the SEM results (Figure S4). The SEM analysis also revealed the formation of a distinct bilayer structure in the T20 system, with globular bacteria enriched in the outer layer and rod-shaped and filamentous bacteria concentrated in the inner layer. Similar to the results of Sekiguchi et al.[61] that observed microbial aggregates, which comprise the methanogens' dominant core and a bacteria-dominant surface.

3.4.2. Microbial Community Structure. To provide an accurate comparison among different digesters at various stages, qPCR was adopted to understand the absolute abundance of microorganisms in the suspension and on the carrier. As shown in Figure 5, the abundance of suspended microorganisms in the carrier-added group was higher than that in the control group. During stage II, gene copy numbers of bacteria and archaea in T20 sludge samples were approximately 127% and 135% higher than that in D20, respectively, and significantly higher than that in the control group ($p < 0.05$). This indicated that carriers created a favorable environment, fostering microbial growth and leading to substantial active biomass retention within the digestive system. The gene copy numbers of bacteria and archaea was further increased under the influence of microaeration,

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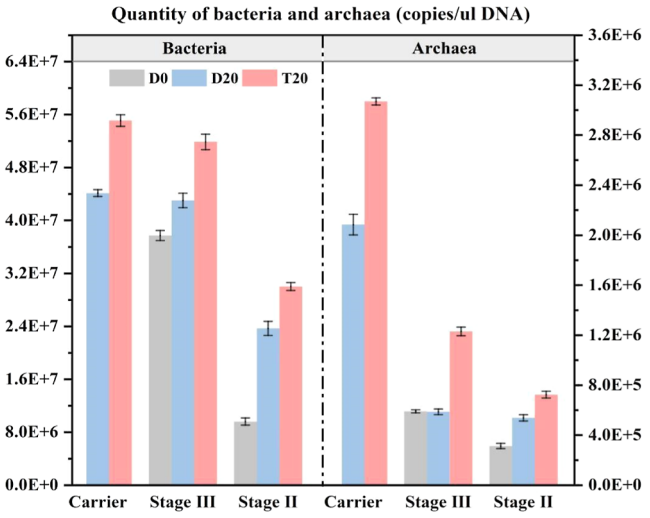


Figure 5. Absolute abundance of bacteria and archaea in the suspension and on the carriers.

indicating that microaeration contributed to the proliferation of microorganisms. Oxygen acts as the terminal electron acceptor during aerobic respiration, enabling microorganisms to generate more energy and grow more efficiently. The observed increase in the ATP content also corroborated the activity of microbial growth and reproduction (Figure 3).

Examination of the absolute gene numbers of microorganisms on the carrier revealed that the gene copy numbers of bacteria and archaea on the conductive carriers were also higher than those on the polyurethane carriers. This difference arose from the ability of conductive carriers to facilitate electron transfer and microbial communication, enhancing metabolic activity and microbial growth.[62] Furthermore, the carrier's porous structure resulted in a higher level of attached microorganisms on its surface compared to suspended microorganisms. This structure effectively trapped biomass, providing a large-surface area for biofilm accumulation, and shielded the biofilm from fluid shear and collisions by offering sheltered anchoring points (Figure S5).[63] Additionally, the microaeration condition led to an increase in EPS secreting, which further accumulates microbial cells on the carriers' surface. Unfortunately, to maintain the original biomass and ensure the sealing of the reactor, microbial analysis on the carriers was only conducted at the end of the entire experiment.

The combination of microaeration and different carriers not only enriched functional microorganisms but also reshaped the microbial community (Figure 6). Phylum level identification based on 16S rRNA high-throughput sequencing analysis revealed that all samples were dominated by *Bacteroidota*, *Firmicutes*, *Actinobacteriota*, *Patescibacteria*, *Synergistota*, *and Chloroflexi* (Figure 6a). Compared with the samples from stage II (without aeration), the relative abundance of some facultative bacteria (i.e., hydrolytic and fermentative bacteria) showed a significant increase after microaeration (stage III). In fact, the first three major bacterial phyla are facultative bacteria, which usually exist in microanaerobic fermentation systems. Among them, *Bacteroidota* is the first dominant phylum, which is an important microbial component in biogas fermentation. *Bacteroidota* is mainly involved in the hydrolysis of polysaccharides and proteins, which is commonly found in the AD system by using FW as a substrate.[64] Compared with stage II, the abundance of *Bacteroidota* in D0 and D20 increased by 34.6% and 26.8%, respectively, while it decreased by about 6.33% in T20. It shows that *Bacteroidota* can survive in D0 and D20 under microaeration environment, and the hydrolysis efficiency of macromolecular organic matter is accelerated due to the stimulation of oxygen. Furthermore, the decrease of *Bacteroidota* in T20 was accompanied by the increase of *Firmicutes* and *Actinobacteria*. *Firmicutes* were able to produce spores for resistance to extreme environments.[65] *Actinobacteria* play a key role in the degradation of organic matter, which can convert organic matters into available substrates for methane production.[66] After microaeration treatment, the relative abundance of *Actinobacteriota* in the T20 group was 42.8% and 21.3% higher than that of D0 and D20. The relative lower abundance of *Firmicutes* and *Actinobacteria* in D0 and D20 demonstrates that the microaeration may have a certain inhibitory effect to them. Previous studies have also found that the increase in *Actinobacteria* abundance is conducive to sludge aggregation and tends to form a stable agglomerated particle structure, which is beneficial to prevent the infiltration of excess O_2 and lead to the toxic effect of reactive oxygen on methanogens,[24] and also corresponds to the microbial aggregates observed in Section 3.4.1. *Chloroflexi* is a filamentous microorganism in the AD system, which can hydrolyze glucose, amino acids, and produce VFA.[67] The relative abundance of *Chloroflexi* in D0, D20, and T20 was 3.41, 4.5, and 3.74% in stage III, which was 1.6, 2, and 1.5 times higher than that in stage II. This finding further validated that introducing microaeration contributed to the enrichment of hydrolytic acidifying bacteria.

The microbial community structure on the carrier was close to that of the suspended sludge. More significant enrichment of *Synergistota* was observed on the carrier to the control groups without carriers. *Synergistota* are obligate anaerobic bacteria, which can accelerate the conversion of VFA into acetate.[68] It was clear that the relative abundance of *Synergistota* was significantly reduced in the control group by the microaeration, whereas it was significantly enriched in the carrier group, and the abundance of *Synergistota* was relatively higher on the T20 carrier, which was 37.0% higher than that of the D20 carrier. This verifies the role of carriers in toxicity "shielding" for strict anaerobic bacteria.[69] In addition, higher abundances of *Fermentibacterota* and *Elusimicrobiota* were also observed on the carrier, which indicates that the biological community structure on the surface of the carrier was more abundant, and thus a more stable ecosystem may achieve.

Changes in the archaeal community were observed in suspended slurry, and the results are shown in Figure 6b. It can be seen from the figure that the dominant archaea at the genus level in the suspension form include *Methanosaeta*, *Methanosarcina*, *unclassified_d__Archaea*, and *Methanolinea*, but their percentage varied among different digesters. Notably, the archaeal community structure in D0 was reshaped after the introduction of microaeration. The dominant genus of methanogens in the second stage of the D0 reactors was *Methanosaeta*, with a relative abundance of 69.95%. The abundance of *Methanosarcina* increased significantly following the introduction of microaeration, accompanied by a decrease in *Methanosaeta*. It can be seen that *Methanosaeta* is highly sensitive to ROS generated by microaeration. Furthermore, the decrease in *Methanosaeta* abundance may be attributed to reduced acetate levels as *Methanosaeta* is an obligate acetoclastic methanogen that exclusively uses acetate as its

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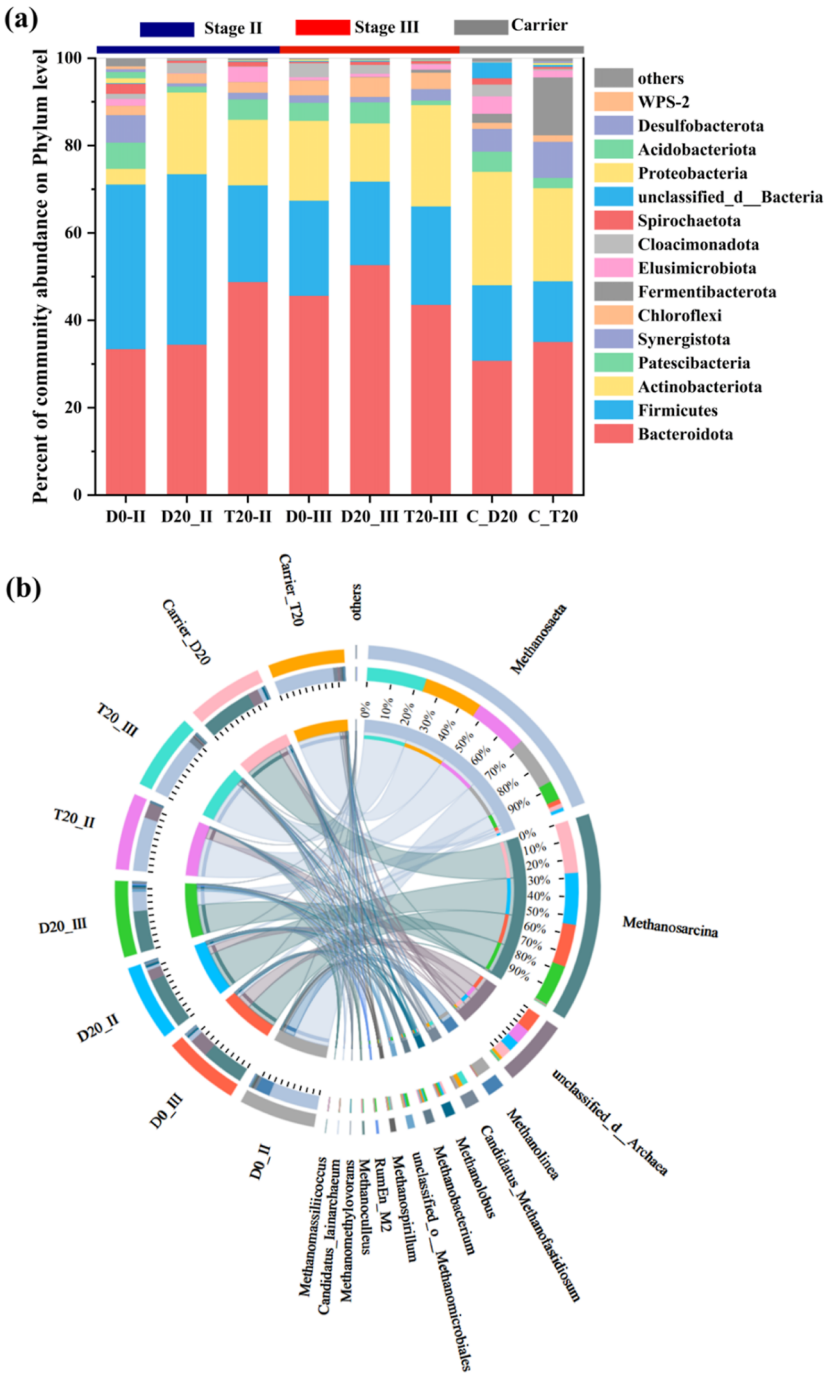


Figure 6. Relative abundance of bacteria at the phylum level (a) and archaea at the genus level (b) in carrier and sludge samples.

sole substrate.[9,70] More facultative bacteria may promote CO_2/H_2 production instead of acetate production. *Methanosarcina*, which utilizes both acetate and CO_2/H_2 for methane formation, has been reported to exhibit strong tolerance to external environmental stress.[71] Yu et al.[12] also found that the growth of *Methanosaeta* was related to the microaerobic volume of the anaerobic system, and the relative abundance of *Methanosaeta* was extremely low under high microaerobic conditions. Compared with D0 group, the archaeal community structure of the carrier group was not affected by microaeration, and the relative abundance of *Methanosaeta* in the carrier group D20 and T20 increased by 84.9 and 9.8%, respectively, indicating that microaeration was beneficial to the growth of *Methanosaeta* in the carrier group.

Meanwhile, the comparison of microbial structures on the two carriers revealed that *Methanosaeta* was the first dominant bacteria in T20, while the relative abundance of *Methanosarcina* was even higher in the reactor with D20, becoming the second dominant genus after *Methanosaeta*, which was consistent with the microbial structure of its suspended sludge sample. This variation may stem from differences in the nature and composition of the carriers, which likely influence the enrichment of functional microbes in distinct ways. The archaea in D20 are centered on *Methanosarcina*, which maintained the stability of the archaeal community through

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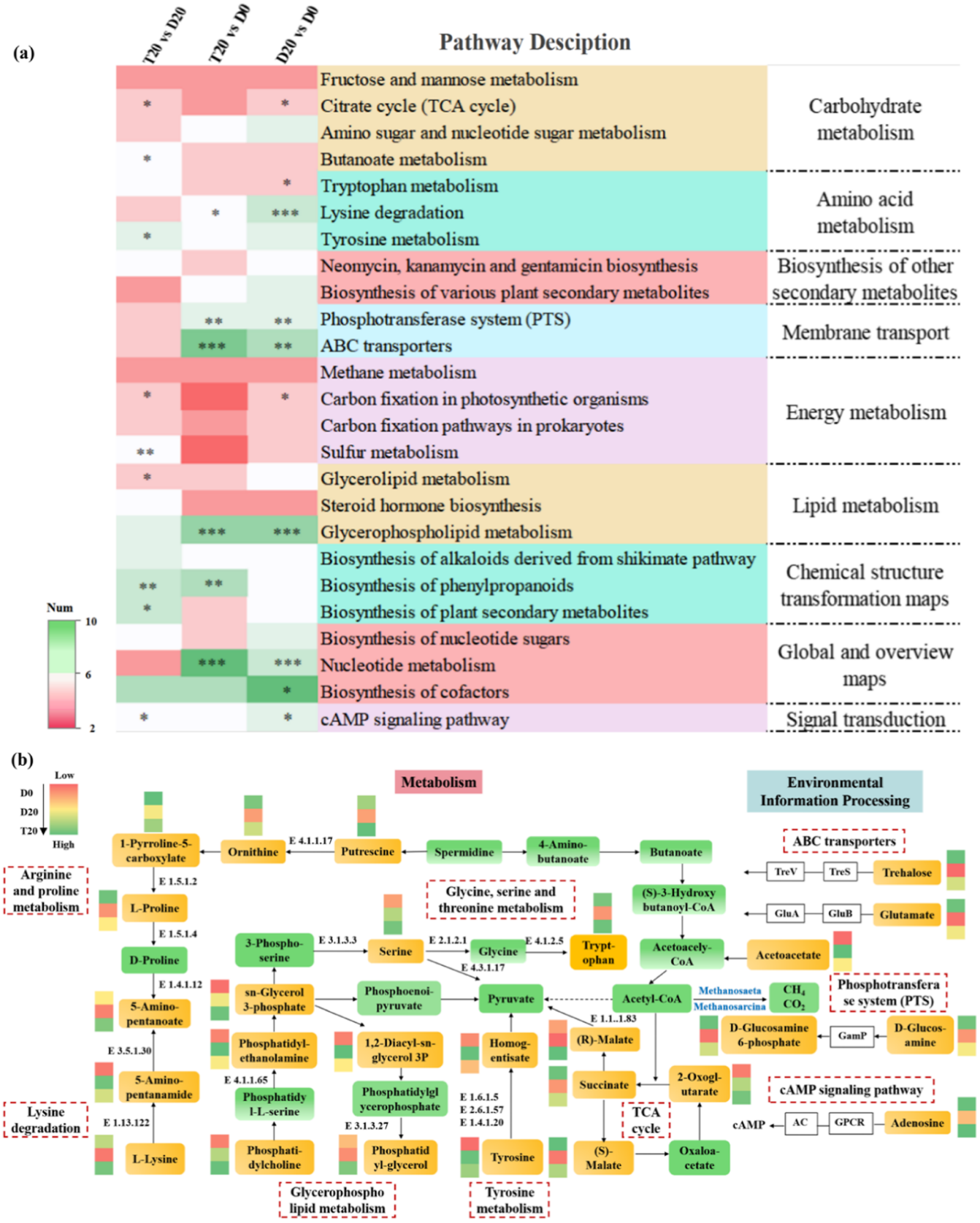


Figure 7. Metabolomics analysis [(a) Top 25 KEGG pathways enriched with the highest number of differential metabolites. (b) Prediction of metabolic pathways of characteristic differential metabolites. The color chart represents the content change of metabolic components in each reactor of the microaeration process. The orange and green blocks represent differential metabolites and intermediates, respectively. Red dashed blocks represent different metabolic pathways. ***$P < 0.001$, **$P < 0.01$, and *$P < 0.05$ indicate significant difference.]

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the secretion of antioxidant enzymes and exhibited strong oxygen tolerance.[52] For digester T20, the conductivity of the carrier may have facilitated the enrichment potential to *Methanosaeta* with membrane-bound cytochromes, allowing direct electron acceptance through the carrier and enhancing resistance to external oxidative stress.[72,73] In addition, the enrichment of hydrogenotrophic methanogens (such as *Methanospirillum* and *Methanobacterium*) was also observed in D20 and T20, which can produce CH_4 by reducing CO_2 with H_2 or formate as an electron donor. Li et al.[74] showed that *Methanospirillum* was replaced by *Methanoculleus* when the reactor became unstable, indicating that the introduction of microaeration did not compromise the stability of the AD system in the carrier group.

3.5. Metabolomic Analysis. *3.5.1. Discriminating Metabolite Analysis.* In order to deeply understand the potential mechanism, untargeted metabolomics analyses were used for samples in the microaeration stages of the three reactors. A total of 6111 mass spectra peaks were detected in 18 samples, corresponding to 696 metabolites. The principal component analysis and partial least-squares discriminant analysis (PLS-DA) revealed significant differences between the different groups (Figure S6). Based on the VIP and p values of the model OPLS-DA, the Student's *t*-test was applied for discriminating the samples from D0 and D20, D0 and T20, and D20 and T20 to identify the differential metabolites, respectively (setting VIP > 1, $p < 0.05$, FC > 1, or FC < 1). A Venn diagram was employed to identify significant differential metabolites among different digestive groups (Figure S7), uncovering 161 unique metabolites with 88 metabolites common across all control groups.

3.5.2. Metabolic Pathway Analysis. The diverse metabolites coordinate with each other to perform biological functions during AD. Analyzing and annotating pathways can provide insights into their functions. Thus, these key metabolites were annotated using the Kyoto Encyclopedia of Genes and Genomes (KEGG) database to determine the associated pathways, further elucidating the mechanisms. The top 25 KEGG pathways with the highest degree of enrichment were selected from the three comparison groups, indicating that the metabolic pathways of the digestion samples primarily involved two main functions: metabolism and environmental information processing (Figure 7a). These pathways were mainly concentrated in carbohydrate metabolism, amino acid metabolism, energy metabolism, and lipid metabolism with some involvement in the biosynthesis of other secondary metabolites, membrane transport, and signal transduction, among others.

The methanogenic pathways were directly or indirectly associated with the central metabolic pathways. Fluctuations in these pathways were influenced by changes in the diverse metabolites. This study constructed a metabolic network of differential metabolites based on KEGG database pathways to comprehensively investigate metabolism (Figure 7b). Summarizing significantly enriched pathways revealed that D0, D20, and T20 samples primarily featured glycerophospholipid metabolism, amino acid metabolism, and signal transduction pathways, focusing subsequent analysis on these pathways.

Microorganisms can produce sn-Glycerol 3-phosphate through the glycerophospholipid metabolism, which is converted into 3-phosphoserine to enter the glycine, serine, and threonine metabolism pathway or converted into phosphoenolpyruvate to participate in the synthesis and metabolism of phosphatidylglycerol or amino acid (e.g., tyrosine, proline, and lysine). Glycerophospholipid, as the major lipid part of the cell membranes, has a variety of biological functions, including cell proliferation, differentiation, and apoptosis. Phosphatidylethanolamines (PEs), phosphatidylcholine (PCs), and lysophosphatidylcholine (LysoPCs) are all types of glycerophospholipids and their derivatives. LysoPCs, formed by the partial hydrolysis of PCs, are considered fatty acid oxidation products and can affect the lipid content in tissues. Increased oxidative stress may disrupt lipid metabolism and cell membrane synthesis.[75] Choline, which is a main component of PCs, is essential for cell membrane phospholipids. Elevated choline levels can lead to oxidative modification of membrane phospholipids, disrupting membrane structure and causing component accumulation.[76] LysoPCs and choline are involved in choline metabolism and glycerophospholipid metabolism pathways. In the study, glycerophospholipid derivatives like LysoPC (20:1(11Z)/0:0) and LysoPC (15:0/0:0) were found to be reduced in the T20 group (Supporting Information statement), indicating a potential regulation of cellular oxidative stress by the conductive carrier, resulting in decreased LysoPC levels. Furthermore, levels of PCs and PEs were also reduced in T20 cells following microaeration treatment. PCs are essential components of cell membranes, while PEs serve as precursors to PCs and are involved in cell membrane turnover. Glycerophospholipid derivatives and choline were downregulated in the T20 group compared with the control, indicating that T20 can resist oxidative stress by attenuating the oxidation of fatty acids, promoting cytokine secretion, and accelerating the normal biosynthesis of cell membranes, which is another reason T20 may be resistant to ROS.

Amino acids are mainly derived from the degradation of proteins and play an important role in biological activities, providing energy and substances for many biosynthesis processes in cells. It is worth mentioning that microaeration treatment changed the metabolism of several amino acids, including tyrosine, tryptophan, serine, lysine, and proline. For example, putrescine, ornithine, 1-pyrroline-5-carboxylate, and proline involved in the proline metabolic pathway were significantly altered in the D0 group compared with the T20 group. The highest levels of intermediates were observed, indicating that microaeration treatment may interfere with the formation of intermediate products in D0 (without carrier). Destructive accumulation or metabolism of intermediate products may affect the oxidative equilibrium and homeostasis of energy metabolism.[76] In contrast, the levels of relevant intermediate metabolites were significantly reversed in the carrier group (D20 and T20), reflecting the beneficial function of the carrier. In AD, amino acid metabolism can also be decomposed and utilized by some bacterial groups to produce energy and a series of intermediate metabolites, thus facilitating decomposition and conversion of organic matters. Butanoate can be transformed to acetyl-CoA, which can be further converted to CH_4 to produce ATP.[77] Succinate and (S)-Malate were intermediates in the TCA cycle, which were significantly reduced in the conductive carrier (T20) group. TCA cycle was the final terminal pathway for carbohydrates, lipids, and proteins, and the downregulation of intermediate products indicates that nutrients are rapidly consumed. The downregulation of its intermediate metabolites in T20 indicated that conductive carriers may accelerate nutrient

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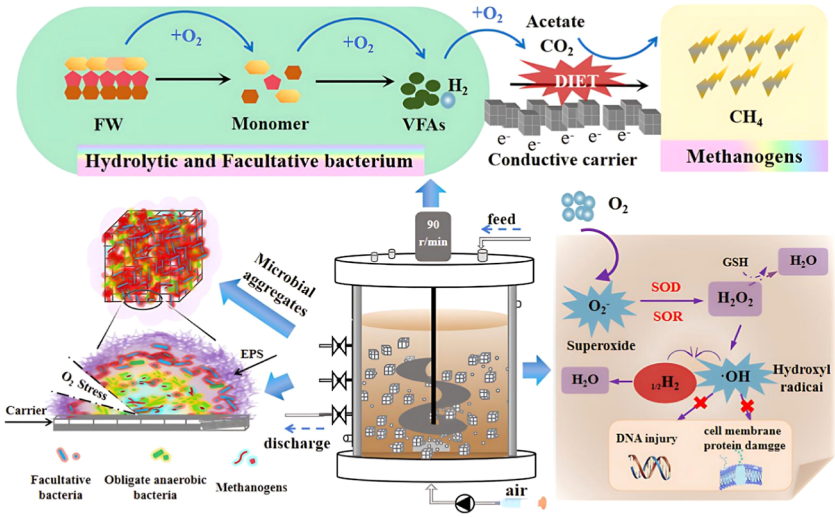


Figure 8. Potential mechanism of combined microaeration and conductive carrier in enhancing the AD performance.

conversion, resulting in the production of more energy and methanogenic precursors.[78]

ABC transporter proteins, c AMP, and the phosphotransferase system (PTS) play an important role in regulating cellular function, metabolism. ABC transporter proteins are associated with metabolic support functions, such as protein transport and ATP transfer. The upregulation of intracellular ABC transporter proteins in T20 indicated that microaeration combined with conductive carriers enhanced ATP transport across bacterial membranes and facilitated the utilization of nutrients by microorganisms.[79] c AMP is an intracellular second messenger responsible for intracellular signal transduction to trigger physiological changes, and it enrichment can activate certain cellular and increase metabolic activity. The PTS system can recognize intracellular and extracellular related signals and convert such signals into the regulation of the physiological activity of the target substance by means of protein interactions.[80] Related studies have also shown that PTS also has functions such as coordinating the metabolic balance of carbon and nitrogen and mediating the stress response.[81] The above pathways were effectively modulated to produce multiple responses to microaeration stress stimulation, enhancing microbial resistance and adaptation to environmental stresses.

4. ENVIRONMENTAL IMPLICATIONS

The combined microaeration and conductive carrier system presented stable digestive performance and better tolerance to ROS. The main reasons for the better performance are as follows: (1) the conductive carrier promoted the conversion of O_2 to ROS and strengthened the hydrolysis of organic matter. In other words, higher level of ROS and better hydrolysis; (2) the conductive carrier led to a good communication and transfer of electrons between bacteria and archaea, which ensured metabolic balance, facilitated the DIET process among microorganisms, made the metabolism of both bacteria and archaea more active, and accelerated the conversion of organic matter to methane; (3) microorganisms formed clusters on biological carrier surfaces, and microaeration boosted the secretion of EPS of microorganisms, which further promoted the formation of aggregate structure and provides a sheltered environment for microorganisms, resulting in better antioxidant capacity of microorganisms, especially for strict anaerobes (Figure 8). Additionally, microaeration stimulated facultative bacteria growth (e.g., *Bacteroidota* and *Actinobacteriota*) that rapidly consume O_2, creating an anaerobic niche for methane production. Methanogens such as *Methanosarcina* maintained archaeal community stability through antioxidant enzyme secretion (e.g., SOD and SOR) and strong oxygen tolerance.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsestengg.4c00298.

Specific methane yield at different filling ratios in the reactor; live/dead bacteria images of sludge collected from D0, D20, and T20 in stage III; analysis results of green (bacteria) and red (archaea) fluorescence signals in the two sets of images using software ImageJ; morphological characteristics of digested sludge and filler surface; polyurethane carrier and conductive carrier; statistical analysis of metabolites extracted from D0III, D20III, and T20III samples; volcano plot and Venn diagram of differential metabolites in stage III; classification of the differential metabolites at the "superclass" level of the HMDB database; characteristics of FW and inoculated sludge; details of the used polyurethane foam carrier and conductive carrier; oligonucleotide probes used in this assay, target groups of each of them, and hybridization conditions; and EPS content at the microaeration stage of D0, D20, and T20 (PDF)

Details of differences between different groups (XLSX)

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Notes

The authors declare no competing financial interest.

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